Chapter 5 Critical Thinking Questions (page 143: #2 - #9 and #11):

2. A method declaration is the blueprint that outlines what the method accepts returns and who can access it, before the { }. The method body, found inside the { } contains the actual code execution and logic that determines how the method performs a task.

3. The first word of a method is the access modifier that determines the access level. For example: public, private, protected…

4. Other words that have the same meaning are access-level modifier and visibility modifier.

5.

- Var1 is local to method1()
- Var2 is local to the for loop only
- Var3 is local to the main method
- var4 is local to the for loop only

6. a) public static int getVowels(String word)
b) public static int extractDigit(int number)
c) public static String insertString(String word, int number)

7. a) Methods are distinguished by their method signature (the method name and set parameters).
b) Two methods in the same class can have the same name as long as they have different parameters. This is referred to as method overloading.

8. a) Return statements are used to send values back to methods that called them.
b) Only one value.
c) Methods that return values declare the data type that will be returned, in the declaration (e.g. int, double, String). If a method doesn't return a value, it just uses "void".

9. The method doSomething() returns the integer 5, but the first time it is called, the code doesn't store or even use the returned value. It's not considered a compilation error but the statement is completely unnecessary. The second call properly stores the returned value.

11. a) True
b) False: The method call does not include the whole method declaration, only the name and its parameters.
c) False: Void methods do return values.
d) False: The access modifier only controls from where the method can be accessed, not what data type the method must return.
e) True
f) False: Method parameters are enclosed by (), the body is enclosed by {}.

g) False: Access to local variables is limited to the method where they are declared.

h) False: Java strictly uses pass-by-value for all argument passing.

i) False: Method overloading is when multiple methods have the same name, not when an app has too many methods.

j) True

k) False: the data types of a method's parameters are part of the method signature, not the precondition.

l) False: The postcondition is entirely unrelated to what the method does, the way that the method accomplishes its task is outlined in the method body.